

The Hidden Nest – Official Solution

Challenge Overview

Challenge Name: The Hidden Nest

Difficulty: Beginner to Intermediate

Estimated Solve Time: 1–3 Hours

Category: Steganography / Forensics / Cryptography

Starting File: welcome.jpg

Flag Format: FLAG{...}

Description: Everything you need is already inside the file.

Step 1: Extract EXIF Metadata

Inspect the image metadata using ExifTool:

```
exiftool welcome.jpg
```

Base64 string found in the Comment field:

```
cGFzc3dvcmQ6IE5lc3RQNDU3IQpIaw500iBBIHNlY3JldCBpcyBuZXN0ZWQgYmVuZWZ0aCB0aGUgc3VyZmFjZS4=
```

Decode the Base64 string:

```
echo "cGFzc3dvcmQ6IE5lc3RQNDU3IQpIaw500iBBIHNlY3JldCBpcyBuZXN0ZWQgYmVuZWZ0aCB0aGUgc3VyZmFjZS4=" | base64 -d
```

password: NestP457!

Hint: A secret is nested beneath the surface.

Step 2: Extract the Hidden ZIP

```
binwalk welcome.jpg
```

```
binwalk -e welcome.jpg
```

Alternatively:

```
unzip welcome.jpg
```

Extracted files:

```
note.txt
fake_flag.txt
audio.wav
```

Step 3: Identify the Fake Flag

```
cat fake_flag.txt
```

```
FLAG{N3st_F0und_But_EmptY}
```

This is a decoy flag.

Step 4: Decode note.txt

```
cat note.txt
```

```
.oidua eht ni neddiH si galf laeR  
!lrXjorepraF si drowssaP dnoceS
```

Reverse the text:

```
rev note.txt
```

Real flag is hidden in the audio.
Second Password is FareprojXrl!

Apply ROT13:

```
echo "FareprojXrl" | tr 'A-Za-z' 'N-ZA-Mn-za-m'
```

SparrowKey

Second Password: SparrowKey
The real flag is hidden in the audio file.

Step 5: Decode the Morse Code Audio

Analyzing audio.wav reveals the Morse message:

KEY PHOENIX STOP ENCODED TFBHR3tTV1xMRzFRT2tsUVQwRlVTVE09

XOR Key: PHOENIX

Encoded String: TFBHR3tTV1xMRzFRT2tsUVQwRlVTVE09

Step 6: Final Decoding

Perform the following operations in order:

1. XOR with key PHOENIX
2. Base64 Decode
3. Apply ROT13

Python script:

```
import base64
```

```
encoded = "TFBHR3tTV1xMRzFRT2tsUVQwRlVTVE09"  
key = "PHOENIX"
```

```
# XOR decryption  
xor_result = ''.join(  
    chr(ord(c) ^ ord(key[i % len(key)]))  
    for i, c in enumerate(encoded)  
)
```

```
# Base64 decode
decoded = base64.b64decode(xor_result).decode()

# Apply ROT13
rot13 = decoded.translate(
    str.maketrans(
        "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz",
        "NOPQRSTUVWXYZABCDEFGHIJKLMnopqrstuvwxyzabcdefghijklm"
    )
)

print(rot13)
```

Final Flag

FLAG{SWEET_NEST_DISCOVERED}

Tools Required

ExifTool
Binwalk
Strings
CyberChef
Audacity or Morse Decoder
Python 3